

Delhi Public School

Subject: Physics | Grade: 12th | Class: 12th

Time: 1 Hour

Maximum Marks: 30

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. Read all questions carefully before answering.
2. Write neatly and legibly.

Section A

This section contains multiple-choice questions, true/false assertions, and fill-in-the-blank questions. Each question carries 1 mark.

Q1. What is the electric potential due to a point charge 'q' at a distance 'r' from it in vacuum? [Easy] [1]

- a. $q / (4 \pi \epsilon_0 r^2)$
- b. $q / (4 \pi \epsilon_0 r)$
- c. $q^2 / (4 \pi \epsilon_0 r)$
- d. $q / (4 \pi \epsilon_0 r^3)$

Expected Answer: $q / (4 \pi \epsilon_0 r)$

Q2. If the current flowing through a copper wire is doubled, what happens to the drift velocity of the free electrons inside the wire (assuming temperature remains constant)? [Moderate] [1]

- a. It remains unchanged.
- b. It becomes halved.
- c. It is doubled.
- d. It becomes four times.

Expected Answer: It is doubled.

Q3. Which of the following is the correct SI unit of magnetic dipole moment? [Moderate] [1]

- a. Ampere-meter
- b. Ampere-meter square
- c. Tesla-meter / Ampere
- d. Tesla-meter square

Expected Answer: Ampere-meter square

Q4. In a purely capacitive AC circuit, what is the phase relationship between the alternating voltage across the capacitor and the current passing through it? [Hard] [1]

- a. Voltage and current are in phase.
- b. Voltage leads current by $\pi / 2$ radians.
- c. Current leads voltage by $\pi / 2$ radians.
- d. Voltage leads current by π radians.

Expected Answer: Current leads voltage by $\pi / 2$ radians.

Q5. Which of the following expressions correctly represents the energy of a photon of wavelength λ ? [Easy] [1]

- a. $h \cdot c / \lambda$
- b. $h \cdot \lambda / c$
- c. $h \cdot c \cdot \lambda$
- d. $e \cdot V$

Expected Answer: $h \cdot c / \lambda$

Q6. Electrostatic field lines always form continuous closed loops starting from positive charges and ending at negative charges. [Easy] [1]

Expected Answer: False

Q7. The resistivity of a semiconductor material decreases with an increase in its temperature. [Moderate] [1]

Expected Answer: True

Q8. According to Gauss's Law for Magnetism, the net magnetic flux through any closed surface is always zero. [Moderate] [1]

Expected Answer: True

Q9. A step-up transformer can be used to increase the net electrical power output of an alternating current source. [Easy] [1]

Expected Answer: False

Q10. According to the de Broglie hypothesis, a heavier particle will always have a longer de Broglie wavelength than a lighter particle moving with the same velocity. [Hard] [1]

Expected Answer: False

Q11. The phenomenon of splitting of white light into its component colors when passing through a prism is called _____. [Easy] [1]

Expected Answer: dispersion

Q12. The focal length of a plane mirror is _____. [Moderate] [1]

Expected Answer: infinite

Q13. When a p-n junction diode is forward biased, the width of the depletion layer _____. [Moderate] [1]

Expected Answer: decreases

Q14. When a dielectric slab of dielectric constant K is completely inserted between the plates of an isolated charged capacitor, its capacitance _____. [Moderate] [1]

Expected Answer: increases

Q15. The minimum energy required by an electron to escape from a metal surface is called its _____. [Hard] [1]

Expected Answer: work function

Section B

This section contains short-answer questions. Each question carries 2 marks.

Q16. Define an equipotential surface. Write any two important properties of an equipotential surface. [Easy] [2]

Expected Answer: An equipotential surface is a surface over which the electrostatic potential remains constant at every point. Properties: 1) No work is done in moving a charge from one point to another on this surface. 2) Electric field lines are always normal (perpendicular) to the equipotential surface at every point.

Q17. State Ohm's law. Draw a representative graph showing the variation of current with voltage for (i) Ohmic conductors, and (ii) Non-ohmic devices (like a diode). [Moderate] [2]

Expected Answer: Ohm's Law states that the current flowing through a conductor is directly proportional to the potential difference across its ends, provided physical conditions like temperature and pressure remain constant ($V = IR$). Ohmic graph is a straight line passing through the origin. Non-ohmic graph is non-linear (curved), indicating variable resistance.

Q18. Two point charges of +2 microcoulombs and -2 microcoulombs are kept separated by a distance of 10 cm in vacuum. Determine the net electric potential at the midpoint of the line joining these two charges. [Hard] [2]

Expected Answer: At the midpoint, the distance of each charge is $r = 5 \text{ cm} = 0.05 \text{ m}$. The potential due to the positive charge is $V_1 = k \cdot q / r$ and the potential due to the negative charge is $V_2 = k \cdot (-q) / r$. Since the charges are equal and opposite, and equidistant from the center, $V_{\text{net}} = V_1 + V_2 = k/r \cdot (q - q) = 0 \text{ Volts}$.

Section C

This section contains long-answer questions. Each question carries 3 marks.

Q19. State Biot-Savart Law and write its mathematical expression in vector form. Use this law to derive an expression for the magnetic field at the center of a circular current-carrying loop of radius R . [Moderate] [3]

Expected Answer: Biot-Savart Law states that $dB = (\mu_0 / 4 \cdot \pi) \cdot (I \cdot dl \times r) / r^3$. For a circular current-carrying loop of radius R , the angle between the current element dl and position vector r at the center is 90 degrees. Thus, $dB = (\mu_0 / 4 \cdot \pi) \cdot (I \cdot dl \cdot \sin(90)) / R^2$. Integrating this around the loop (integral of $dl = 2 \cdot \pi \cdot R$), we get the total magnetic field $B = \text{integral}(dB) = (\mu_0 \cdot I \cdot 2 \cdot \pi \cdot R) / (4 \cdot \pi \cdot R^2) = (\mu_0 \cdot I) / (2 \cdot R)$.

Q20. State Faraday's Laws of Electromagnetic Induction. A square loop of side 10 cm and resistance 0.5 ohm is placed vertically in an east-west plane. A uniform magnetic field of 0.10 T is set up across the plane in the north-east direction. If the magnetic field is decreased steadily to zero in 0.70 seconds, calculate the magnitude of the induced electromotive force (emf) and current during this time. [Hard] [3]

Expected Answer: Faraday's Laws: 1) Whenever magnetic flux linked with a circuit changes, an emf is induced. 2) Induced emf is $e = -d(\Phi)/dt$. Given: Area $A = 0.1 \cdot 0.1 = 0.01 \text{ m}^2$. The angle θ between the normal to the east-west plane (pointing North) and the magnetic field (North-East) is 45 degrees. Initial flux $\Phi_i = B \cdot A \cdot \cos(45) = 0.10 \cdot 0.01 \cdot (1/\sqrt{2}) = 7.07 \cdot 10^{-4} \text{ Wb}$. Final flux $\Phi_f = 0$. Change in flux $d(\Phi) = 7.07 \cdot 10^{-4} \text{ Wb}$. Induced emf $e = d(\Phi) / dt = 7.07 \cdot 10^{-4} / 0.70 = 1.01 \cdot 10^{-3} \text{ V}$ (or $\sim 1 \text{ mV}$). Induced current $I = e / R = 1.01 \cdot 10^{-3} / 0.5 = 2.02 \cdot 10^{-3} \text{ A}$ (or $\sim 2 \text{ mA}$).

Q21. Draw a neat labeled ray diagram showing the image formation of a distant object by a refracting telescope in normal adjustment. Write the formulas for its magnifying power and the overall length of the telescope tube in this adjustment. [Moderate] [3]

Expected Answer: The ray diagram must show: 1) Parallel rays from a distant object incident on the objective lens, converging to its focus f_o . 2) This real image acts as an object at the focus f_e of the eyepiece. 3) The final virtual image is formed at infinity. Formula for Magnifying Power: $m = -f_o / f_e$. Formula for Tube Length: $L = f_o + f_e$, where f_o is the focal length of the objective lens and f_e is the focal length of the eyepiece lens.

— End of Question Paper —